

Hockey

Note: The Sports Performance Coach will focus on position players within hockey and not specifically on the goaltending position. In the case of a performance program with hockey players, position players and goalies will receive tremendous benefits by following the application guidelines and techniques presented subsequently. The goaltending position is unique within the sport; however, the same basic fundamentals should be considered when analyzing movement patterns, the demand of the sport/position, and enhancing performance.

If an individual needs to modify the model based on the athlete's needs, then it is very appropriate to do so based on the assessment process, the position's needs, and the goals set for the athlete. The Sports Performance Coach will focus on "off-ice" performance training to decrease injury and increase "on-ice" performance.

Demands of the Sport	
General Information	Specific Information (Bangsbo et al., 2006)
• Activity characterized by high-intensity intermittent skating, rapid changes in velocity and duration • Aerobic and anaerobic energy systems are important to cardiorespiratory conditioning • Movement is primarily skating forward without the puck; however, movement is rarely linear for more than a couple of strides. This should not be confused with lateral motion; turns can range from gradual to sharp, and therefore loading of the legs is rarely equal • Change of direction includes stopping, backward skating, and some lateral movement.	• Duration of game: 60 minutes (3 × 20-minute periods) • Player on ice (shift): 15–20 minutes; defensemen typically play more minutes than forwards • Duration of each shift: 30–80 seconds • Average shift length: 39 seconds • Recovery between shifts: 3–4 minutes • Work-to-rest ratio of 1:2 to 1:3 • Brief periods of accelerated sprints while on the ice: 4–7 seconds • Frequent body contact (player-to-player, player-to-boards), which adds additional stress to the body

Suggested Assessments for Hockey

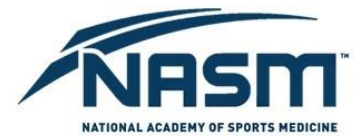
Assessment & Program Strategies: Hockey



Assessment Type	Assessment Method
Transitional Postural Assessments	- Overhead squat test - Single-leg squat test - Split squat - Star excursion balance test
Mobility Assessments (if needed)	- Foot and ankle testing - Lumbo-pelvic-hip complex testing - Shoulder and spine testing
Dynamic Movement Assessments	- Drop jump - Shark skill test - Davies test - Triple crossover hop test
Aerobic test (choose one of the following)	12-minute run test 1.5-mile run test - Beep test
Anaerobic test	300-yard shuttle run
Muscular Endurance Assessments	- Push-up test - Pull-up test - Curl-up test
Muscular Strength Assessments	- Grip strength 1–5 RM bench press 1–5 RM squat 1–5 RM deadlift
Power Assessments	- Vertical jump test (double leg) - Vertical jump test (single leg) - Rotational medicine ball throw test (L & R)
Speed Assessments	40-yard sprint test
Agility Assessments	- T-test 5-10-5 (Pro agility test)

Off-Season Goals and Considerations for Hockey

Assessment & Program Strategies: Hockey



Goals	Considerations
- Physical and mental reconditioning - Reconditioning from the physical contact and additional stress placed on the body - Correcting impaired movement patterns based on results of transitional postural assessments - Improving - Range of motion, joint stability, and muscle strength & endurance - Cardiorespiratory endurance (aerobic/anaerobic) - Total body stabilization, strength, and power - Speed, agility, and quickness - Increasing lean body mass and size - Determine, through assessment, demand of sport analysis, and athlete goals, which plan and program should be created to achieve both the needs and the goals	- Planning and programming will be based on assessment, client goals, and the demand of the sport - A systematic off-season plan will incorporate integrated training, including: - Core training - Balance training - Power training - Speed, agility, and quickness training - Integrated resistance training - Cardiorespiratory training combined with a proper nutrition plan to maximize training and recovery - Opportunities for appropriate rest between workouts to be provided - An off-season training schedule, that works backward from the commencement of the preseason, will be developed, to allow for maximum injury prevention and performance gains without the athlete overtraining - Conditioning for Hockey will be as similar to the game demands as possible - Off-ice conditioning utilizing cardio machines and exercises incorporating ground-reaction forces will be considered when designing a conditioning program

NOTE: Annual programming and periodization is based on elite-level hockey schedules. Lower-level and younger athletes may play shortened seasons and, therefore, will have a longer off-season. Off-season programming should be adjusted in this case to progress the athlete at the proper pace to reach the preseason in optimal condition.

Sample Off-Season OPT™ Hockey Plan (May–August)

Level	Phase	JAN- APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC
		IN- Season	OFF- Season (1)	OFF- Season (2)	OFF- Season (3)	OFF- Season (4)	PRE- Season	IN- Season	IN- Season	IN- Season
Corrective			X							
Stabilization	1		X			X				
Strength	2			X	X	X				
	3									
	4			X						
Power	5				X	X				
	6									

Sample Off-Season Weekly OPT™ Hockey Schedule

Assessment & Program Strategies: Hockey



Month 1: 3 days/week

	MON	TUE	WED	THU	FRI	SAT	SUN
OPT™	Stabilization	OFF	Corrective	OFF	Stabilization	OFF	OFF
Phase	Endurance				Endurance		

Month 2: 4 days/week

	MON	TUE	WED	THU	FRI	SAT	SUN
OPT™	Strength	Corrective	Maximal	OFF	Strength	OFF	OFF
Phase	Endurance		Strength		Endurance		
	Upper Body		Total Body		Lower Body		

Month 3: 5 days/week

	MON	TUE	WED	THU	FRI	SAT	SUN
OPT™	Power	Conditioning	Strength	Conditioning	Power	OFF	OFF
Phase			Endurance				

Month 4: 5 days/week

	MON	TUE	WED	THU	FRI	SAT	SUN
OPT™	Strength	OFF	Stabilization	OFF	Power	OFF	OFF
Phase	Endurance		Endurance				
Other		Conditioning	Conditioning	Conditioning			

Sample Off-Season OPT™ Hockey Programs

Corrective Exercise Training

INHIBIT				Sets	Reps	Time	
Myofascial Rolling							
IT band				1		30 sec	
Adductor				1		30 sec	
Tensor Fascia Latae				1		30 sec	
LENGTHEN				Sets	Reps	Time	
Gastrocnemius stretch				1		30 sec	
Supine biceps femoris stretch				1		30 sec	
Standing adductor stretch				1		30 sec	
Standing psoas stretch				1		30 sec	
ACTIVATE			Sets	Reps	Tempo	Intensity	Rest
Single-leg hip abduction			3	10	4/2/1		0 sec
Single-leg hip adduction			3	10	4/2/1		0 sec
SL bent over cobra			3	10	4/2/1		60 sec
INTEGRATE			Sets	Reps	Tempo	Intensity	Rest
Total body turning step-up, to an overhead press: Single-arm			3	10	4/2/1	60%	60 sec

Phase I: Stabilization Endurance Training

Assessment & Program Strategies: Hockey



WARM-UP			Sets	Reps	Time	
Myofascial Rolling						
Tensor fascia latae			1		30 sec	
Adductors			1		30 sec	
Static Stretching						
Gastrocnemius stretch			1		30 sec	
Kneeling hip flexor stretch			1		30 sec	
Standing biceps femoris stretch			1		30 sec	
Dynamic Stretching		Sets	Reps	Tempo		
Lunge with rotation		1	10	Moderate	0 sec	
Medicine ball woodchop		1	10	Moderate	0 sec	
CORE & BALANCE		Sets	Reps	Tempo	Rest	
Quadruped arm opposite leg raise		3	12	4/2/1	0 sec	
Floor cobra		3	12	4/2/1	0 sec	
Single-leg lift and chop		3	12	4/2/1	60 sec	
PLYOMETRICS		Sets	Reps	Tempo	Rest	
Box jump-down with stabilization		3	8	Hold landing	60 sec	
SPEED/AGILITY/QUICKNESS		Sets	Reps	Tempo	Rest	
Speed ladder: Six exercises		2			60 sec	
RESISTANCE		Sets	Reps	Tempo	Intensity	Rest
Total body	Optional					
Chest	Cable chest press: Alternate arm	3	12	4/2/1	70%	0 sec
Back	Suspension SL row	3	12	4/2/1	70%	0 sec
Shoulders	Single-leg overhead DB press: Single-arm	3	12	4/2/1	70%	0 sec
Legs	Single-leg squat	3	12	4/2/1	70%	90 sec
COOL DOWN						
Myofascial rolling and static stretching						

Phase 2: Strength Endurance Training

Assessment & Program Strategies: Hockey



WARM-UP		Sets	Reps	Time		
Myofascial Rolling						
IT band		1			30 sec	
Piriformis		1			30 sec	
Tensor fascia latae		1			30 sec	
Active Stretching						
Standing adductor stretch		1	10			
Standing hip flexor stretch		1	10			
Floor lat stretch		1	10			
Dynamic Stretching		Sets	Reps	Tempo		
Multiplanar lunge with medicine ball wood chop		1	10	Moderately fast	0 sec	
CORE & BALANCE		Sets	Reps	Tempo	Rest	
Cable lift		2	10	1/1/1	0 sec	
Back extension		2	10	1/1/1	0 sec	
Single-leg squat		2	10	1/1/1	60 sec	
PLYOMETRICS		Sets	Reps	Tempo	Rest	
Repeat squat jumps		3	10	Repeating	0 sec	
Repeat box jumps		3	10	Repeating	60 sec	
SPEED/AGILITY/QUICKNESS		Sets	Reps	Tempo	Rest	
Speed ladder: Six exercises		2			60 sec	
LEFT test		3			45 sec	
RESISTANCE		Sets	Reps	Tempo	Intensity	Rest
Total body	Optional					
Chest	Incline DB press	3	8	2/0/2	80%	0 sec
	Cable chest press		12	4/2/1	50%	
Back	Seated lat pulldown	3	8	2/0/2	80%	0 sec
	Single-leg cable row		12	4/2/1	50%	
Shoulders	Seated overhead DB press		8	2/0/2	80%	0 sec
	Bent over DB combo II		12	4/2/1	50%	
Legs	Barbell squats	3	8	2/0/2	80%	60 sec
	Side lunge to balance		12	4/2/1	50%	
COOL DOWN						
Myofascial rolling and static stretching						

Phase 4: Maximal Strength Training

Assessment & Program Strategies: Hockey



WARM-UP		Sets	Reps	Time
Myofascial Rolling				
IT band		1		30 sec
Piriformis		1		30 sec
Lats/pecs		1		30 sec
Active Stretching				
Standing adductor stretch		1	10	
Standing hip flexor stretch		1	10	
Floor lat stretch		1	10	
Dynamic Stretching		Sets	Reps	Tempo
"World's Greatest Stretch"		1	10	Controlled
CORE & BALANCE		Sets	Reps	Tempo
Cable rotations		2	12	1/1/1
Cable lifts		2	12	1/1/1
Step-up to balance: Frontal plane		2	12	1/1/1
PLYOMETRICS		Sets	Reps	Tempo
No PLYOMETRICS on this day				
SPEED/AGILITY/QUICKNESS		Sets	Reps	Tempo
No SAQ on this day				
RESISTANCE		Sets	Reps	Tempo
Total body	Barbell clean	4	5	Explosive
Chest	Bench press	4	5	1/0/1
Back	Done in total body exercise			
Shoulders	Seated overhead barbell press	4	5	1/0/1
Biceps	Seated DB curl	4	5	1/0/1
Triceps	Supine triceps extensions	4	5	1/0/1
Legs	Done in total body exercise			
COOL DOWN				
Myofascial rolling and static stretching				

Phase 5: Power Training

Assessment & Program Strategies: Hockey



WARM-UP			Sets	Reps	Time	
Myofascial Rolling						
Tensor fascia latae			1		30 sec	
Adductor			1		30 sec	
Thoracic spine			1		30 sec	
Dynamic Warm-up (some athletes may still require static stretching prior to Dynamic warm-up)						
Lateral tube walking			1	10	0 sec	
Prisoner squats			1	10	0 sec	
Lunge with rotation			1	10	0 sec	
Leg swings: Side to side			1	10	0 sec	
MB lift and chop			1	10	0 sec	
MB rotations			1	10	0 sec	
Push-up with rotation			1	10	60 sec	
CORE & BALANCE			Sets	Reps	Tempo	Rest
No CORE & BALANCE (incorporated into Dynamic warm-up)						
PLYOMETRICS			Sets	Reps	Tempo	Rest
No PLYOMETRICS (incorporated into Dynamic warm-up))						
SPEED/AGILITY/QUICKNESS			Sets	Reps	Tempo	Rest
Speed ladder: Six exercises			2			60 sec
T-Drill			4			60 sec
RESISTANCE		Sets	Reps	Tempo	Intensity	Rest
Total body	Deadlift shrug to calf raise	4	4	1/0/1	90%	0 sec
	Push press		8	Explosive	4% of BW	2 min
Chest	Bench press	4	4	1/0/1	90%	0 sec
	Rotational chest press		8	Explosive	4% of BW	2 min
Back	Seated cable row	4	4	1/0/1	90%	0 sec
	Overhead MB throw		8	Explosive	4% of BW	2 min
Shoulders	Done in total body exercises					
Legs	Barbell squats	4	4	1/0/1	90%	0 sec
	Power step-up		8	Explosive	4% of BW	3 min
COOL DOWN						
Myofascial rolling and static stretching						

Preseason Goals and Considerations for Hockey

Assessment & Program Strategies: Hockey



Goals	Considerations
- Correcting impaired movement patterns based on results of preseason assessments - Maintaining: - Physical health (decreases the chance of injury for participation in practice and games, and optimal performance within such activities) - Range of motion, joint stability, and muscle strength and endurance - Cardiorespiratory endurance (aerobic/anaerobic) - Total body stabilization, strength, and powers	- A systematic preseason plan that transitions the athlete from performance enhancement to injury prevention and maintenance of off-season gains will be developed. This transition should be implemented with the consideration that the athlete's sport participation level is increasing dramatically from what they have become used to in the off-season - Proper recovery and conditioning, during this time of increased activity, will be applied - Exercise volume of injury prevention and performance training based on the athlete's increased activity level to be considered

Sample Preseason OPT™ Hockey Plan (September)

Level	Phase	JAN- APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV- DEC
		IN- Season	OFF- Season	OFF- Season	OFF- Season	PRE- Season (1)	IN- Season	IN- Season	IN- Season
Corrective								X	
Stabilization	1							X	
Strength	2							X	
	3								
	4								
Power	5								
	6								

Sample Preseason Weekly OPT™ Hockey Schedule

Month 1

	MON	TUE	WED	THU	FRI	SAT	SUN
OPT™ Phase	Strength Endurance	OFF	Corrective	OFF	Stabilization Endurance	OFF	Corrective
Other		Game		Game		Game	

Sample Preseason OPT™ Hockey Programs

Corrective Exercise Training

Assessment & Program Strategies: Hockey

INHIBIT				Sets	Reps	Time
Myofascial Rolling						
Tensor fascia latae				1		30 sec
IT band				1		30 sec
Adductors				1		30 sec
LENGTHEN				Sets	Reps	Time
Supine biceps femoris stretch				1		30 sec
Kneeling hip flexor stretch				1		30 sec
Standing adductor stretch				1		30 sec
Floor lat stretch				1		30 sec
ACTIVATE		Sets	Reps	Tempo	Intensity	Rest
Side-lying hip adduction		1	15	4/2/1		0 sec
Suspension bridge		1	15	4/2/1		90 sec
INTEGRATE		Sets	Reps	Tempo	Intensity	Rest
Total Body	Squat, curl to overhead press	1	15	4/2/1	60%	90 sec

Phase I: Stabilization Endurance Training

WARM-UP			Sets	Reps	Time	
Myofascial Rolling						
Tensor fascia latae			1		30 sec	
Piriformis			1		30 sec	
Static Stretching						
Standing adductor stretch			1		30 sec	
Kneeling hip flexor stretch			1		30 sec	
Posterior shoulder stretch			1		30 sec	
Dynamic Stretching		Sets	Reps	Tempo		
Lunge with rotation		1	10	Moderate	0 sec	
Medicine ball wood chop		1	10	Moderate	0 sec	
CORE & BALANCE		Sets	Reps		Temp o	
Side iso-ab w/hip abduction		2	15	4/2/1	0 sec	
Floor bridge: Single-leg		2	15	4/2/1	0 sec	
Single-leg throw and catch		2	15	4/2/1	60 sec	
PLYOMETRICS		Sets	Reps	Tempo	Rest	
No PLYOMETRICS due to increased activity						
SPEED/AGILITY/QUICKNESS		Sets	Reps	Distance	Rest	
No SAQ due to increased on-ice conditioning						
RESISTANCE		Sets	Reps	Tempo	Intensity	Rest
Total body	Cable squat to row	2	15	4/2/1	65%	0 sec
Chest	SL suspension chest press	2	15	4/2/1	65%	0 sec
Back	Done in total body exercise					
Shoulders	Single-leg scaption	2	15	4/2/1	65%	0 sec
Legs	Single-leg Romanian deadlift	2	12 ea.	4/2/1	65%	90 sec
COOL DOWN						
Myofascial rolling and static stretching						

Phase 2: Strength Endurance Training

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In-Season Goals and Considerations for Hockey

Assessment & Program Strategies: Hockey



Goals	Considerations
- Maintaining: - Range of motion, joint stability, and muscle strength and endurance - Cardiorespiratory endurance (aerobic/anaerobic) - Total body stabilization, strength, and power - Establishing with the athlete a communication process for monitoring in-season maintenance, and a plan that works based on what the athlete needs, wants, and will have the capability to do focusing on their job playing Hockey	- Assessments and testing to ensure program considerations are being made to prevent overuse injuries and maintain performance levels, will be performed - Impaired movement patterns with exercise strategies being implemented will be attended to - Additional conditioning, for athletes who may not get the same type of on-ice practice or game minutes as others, will be considered - A systematic in-season maintenance plan will be developed that allows the athlete to minimize decreases in performance and chance of injury over a long season with repetitive sport movement patterns and physical contact - The exercise volume of injury prevention, and performance training based on the athlete's increased travel and activity level, will be considered

In-Season OPT™ Hockey Plan (October–April)

Level	Phase	JAN	FEB	MAR	APR	MAY-AUG	SEP	OCT	NOV	DEC
		IN-Season (4)	IN-Season (5)	IN-Season (6)	IN-Season (7)	OFF-Season	PRE-Season	IN-Season (1)	IN-Season (2)	IN-Season (3)
Corrective		X	X	X	X			X	X	X
Stabilization	1	X	X	X	X			X	X	X
Strength	2		X		X			X		X
	3									
	4	X		X					X	
Power	5									
	6									

Sample In-Season Weekly OPT™ Hockey Schedule

Any Month

	MON	TUE	WED	THU	FRI	SAT	SUN
OPT™ Phase	Maximal strength	OFF	Corrective	OFF	Stabilization Endurance	OFF	Corrective
Other		Game		Game		Game	

Sample In-Season OPT™ Hockey Program

Corrective Exercise Training

Assessment & Program Strategies: Hockey



INHIBIT				Sets	Reps	Time
Myofascial Rolling						
Tensor fascia latae				1		30 sec
Adductors				1		30 sec
Thoracic spine				1		30 sec
LENGTHEN				Sets	Reps	Time
Supine biceps femoris stretch				1		30 sec
Kneeling hip flexor stretch				1		30 sec
Standing adductor stretch				1		30 sec
Standing pec stretch				1		30 sec
ACTIVATE		Sets	Reps	Tempo	Intensity	Rest
Side-lying hip adduction		1	15	4/2/1		0 sec
Suspension bridge		1	15	4/2/1		0 sec
Single-leg extension		1	15	4/2/1		90 sec
INTEGRATE		Sets	Reps	Tempo	Intensity	Rest
Total body	Side lunge to overhead press	1	15	4/2/1	60%	90 sec

Phase I: Stabilization Endurance Training

WARM-UP			Sets	Reps	Time	
Myofascial Rolling						
Tensor fascia latae			1		30 sec	
Piriformis			1		30 sec	
Static Stretching						
Gastrocnemius stretch			1		30 sec	
Kneeling hip flexor stretch			1		30 sec	
Posterior shoulder stretch			1		30 sec	
Dynamic Stretching		Sets	Reps	Tempo		
Lunge with rotation		1	10	Moderate	0 sec	
Medicine ball wood chop		1	10	Moderate	0 sec	
CORE & BALANCE		Sets	Reps	Tempo	Rest	
Prone iso-abs		2	15	4/2/1	0 sec	
Bent over SL cobra		2	15	4/2/1	0 sec	
Single-leg balance reach		2	15	4/2/1	60 sec	
PLYOMETRICS, SPEED/AGILITY/QUICKNESS		Sets	Reps	Tempo	Rest	
No PLYOMETRICS or SAQ due to increased activity and on-ice conditioning						
RESISTANCE		Sets	Reps	Tempo	Intensity	Rest
Total body	Lunge, balance to overhead press	2	15	4/2/1	65%	0 sec
Chest	SL suspension chest press	2	15	4/2/1	65%	0 sec
Back	Single-leg cable row	2	15	4/2/1	65%	0 sec
Shoulders	Done in total body exercise					
Legs	Step-up to balance: Transverse plane	2	15	4/2/1	65%	90 sec
COOL DOWN						
Myofascial rolling and static stretching						

Phase 2: Strength Endurance Training

Assessment & Program Strategies: Hockey

WARM-UP		Sets	Reps	Time		
Myofascial Rolling						
Tensor fascia latae		1		30 sec		
Thoracic spine		1		30 sec		
Active Stretching						
Standing adductor stretch		1	10			
Standing hip flexor stretch		1	10			
Floor lat stretch		1	10			
Dynamic Stretching		Sets	Reps	Tempo	Rest	
Multiplanar lunge with medicine ball wood chop		1	10	Moderately fast	0 sec	
CORE & BALANCE						
Cable rotation		2	12	1/1/1	0 sec	
Crunch		2	12	1/1/1	0 sec	
Single-leg Romanian deadlift		2	12	1/1/1	60 sec	
PLYOMETRICS		Sets	Reps	Tempo	Rest	
No PLYOMETRICS because of increased activity						
SPEED/AGILITY/QUICKNESS		Sets	Reps	Distance	Rest	
No SAQ because of on-ice conditioning						
RESISTANCE		Sets	Reps	Tempo	Intensity	Rest
Total body	Cable squat to row	2	12	2/0/2	70%	0 sec
	Step up, balance to overhead press		8	4/2/1		
Chest	Incline bench press	2	12	2/0/2	70%	0 sec
	Push up		8	4/2/1		
Back	Done in total body exercise					
Shoulders	Done in total body exercise					
Biceps	Standing DB curl	2	12	2/0/2	70%	0 sec
Triceps	Standing suspension extension	2	12	4/2/1	70%	0 sec
Legs	Barbell squat	2	12	2/0/2	70%	60 sec
	SL RDL		8	4/2/1		
COOL DOWN						
Myofascial rolling and static stretching						

Phase 4: Maximal Strength Training

Assessment & Program Strategies: Hockey

WARM-UP		Sets	Reps	Time
Myofascial Rolling				
IT band		1		30 sec
Adductors		1		30 sec
Tensor fascia latae		1		30 sec
Active Stretching				
Standing adductor stretch		1	10	
Standing hip flexor stretch		1	10	
Standing pec stretch		1	10	
Floor lat stretch		1	10	
Dynamic Stretching		Sets	Reps	Tempo
"World's Greatest Stretch"		1	10	Controlled
CORE & BALANCE				
Knee-ups		2	12	1/1/1
Back extension		2	12	1/1/1
Step-up to balance		2	12	1/1/1
PLYOMETRICS		Sets	Reps	Tempo
No PLYOMETRICS on this day				
SPEED/AGILITY/QUICKNESS		Sets	Reps	Distance
No SAQ on this day				
RESISTANCE		Sets	Reps	Tempo
Total body	Deadlift shrug to calf raise	4	5	Explosive
Chest	Bench press	4	5	Controlled
Back	Seated cable row	4	5	Controlled
Shoulders	Seated overhead barbell press	4	5	Controlled
Legs	Barbell squat	4	5	Controlled
COOL DOWN				
Myofascial rolling and static stretching				